# 47. Zero Trust Networking Ideas

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**Phase:** Phase 9 — VPN, tunnels, and overlays
**Estimated time:** 1 day
**Slug:** `47-zero-trust`

### Learning objectives

- Move beyond flat VPN trust
- Prefer identity-aware access
- Least privilege paths
- Combine device posture + user identity

### Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

Zero trust: never trust network location alone. Authenticate/authorize continuously.

Practical steps: SSO, short-lived certs, mTLS/mesh, per-service policies, avoid huge flat /16 VPNs with implicit trust.

```
# Still verify paths explicitly
curl -I https://internal.example.com
# without VPN / with identity proxy patterns (org-specific)
```

### Common mistakes

- VPN equals secure forever
- Flat allow-all east-west
- Long-lived shared secrets

### Exercises

Work in `exercises/47-zero-trust/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

**Do not open `solutions/47-zero-trust/` until after an honest attempt.**

### Self-check

1. Zero trust core idea?
2. Risk of flat VPN?

3. Short-lived creds why?

## Further reading

- https://www.nist.gov/publications/zero-trust-architecture